

PROFESSIONAL SUMMARY

AI Engineer with 1+ year of experience building enterprise GenAI systems, multi-agent orchestration platforms, and AI workflow automation solutions within BFSI-focused environments at TCS. Experienced in RAG pipelines, FastAPI backend systems, Azure OpenAI, agentic AI architectures, and real-time AI workflows.

EDUCATION

Heritage Institute of Technology

B.Tech in Information Technology; DGPA: 8.02/10

Kolkata, India

2020 – 2024

EXPERIENCE

Tata Consultancy Services (TCS)

System Engineer

Kolkata, India

Apr 2025 – Present

Enterprise AI Solutions: Designed and developed GenAI-based automation solutions and AI prototypes for BFSI enterprise use cases in collaboration with internal stakeholders and client teams.

AI Test Suite Generator: Built AI-powered Test Suite Generator using Azure OpenAI, RAG pipelines, and prompt engineering techniques to generate 1200+ automation-ready test cases from business requirement documents.

QA Automation Impact: Reduced manual QA effort by approximately 80–90% through automated test generation workflows validated by enterprise QA teams.

Backend Engineering: Developed real-time backend systems using FastAPI, async execution, REST APIs, and WebSockets for streaming AI workflows and orchestration.

Multi-Agent Orchestration: Architected scalable Multi-Agent Orchestrator platform enabling modular agent integration using A2A communication protocols.

AI Workflow Integration: Integrated Azure OpenAI, Groq API, Google ADK, LangChain, and PostgreSQL into unified orchestration workflows and persistent AI systems.

PROJECTS

Multi-Agent Orchestrator Platform: Built scalable orchestration platform for managing communication and execution across multiple AI agents using FastAPI, PostgreSQL, WebSockets, and Azure OpenAI. Implemented async execution, A2A-based communication layers, and accelerator framework to auto-generate A2A-compliant servers from agent scripts.

FinGuard-AI: Developed AI-powered personal finance platform using RAG pipelines and LLMs to analyze financial behavior, generate personalized recommendations, and evaluate spending risks using real-time inference through Groq API.

ACHIEVEMENTS

Runner-Up: TCS Quantum Hackathon for developing innovative technology solutions in collaborative team environment.

Participation: Participated in Google 5-Day Agentic AI Capstone Program focused on modern agentic AI systems and workflows.

SKILLS

Programming Languages: Python, C++

AI / GenAI: LLMs, RAG, Prompt Engineering, Agentic AI, Multi-Agent Systems, MCP, A2A

Frameworks / Tools: Azure OpenAI, LangChain, FastAPI, Google ADK, FastMCP, Groq API

Backend / Infra: REST APIs, WebSockets, PostgreSQL, Async Execution, Real-Time Systems

Concepts: Workflow Automation, Scalable Backend Architecture, AI Orchestration